F00-SR-001
PLANT 33kV HV DISTRIBUTION SWITCHROOM

F00-SW-010
PLANT 33kV HV DISTRIBUTION SWITCHBOARD

B01 B02 B03 B04

36kV 2000A 3~E 31.5kA FOR 3s

630A 630A CLOSE TRIP PR SEL 751 50 50G 27 51 51G NOTE 1 AFD

300/1A 2.5VA 5P20 TEST
300/1A 1VA 10P10 TEST
NOTE 2

B01, B02, B04: 630A 630A CLOSE TRIP PR SEL 751 50 50G 27 51 51G NOTE 1 AFD

B03: 630A 630A CLOSE TRIP PR SEL 751 50 50G 27 51 51G 59 NOTE 1 AFD; PR SEL 787-4 50 51; PR SEL 787-4 50 51

500/1A 2.5VA 5P20 TEST
300/1A 1VA 10P10 TEST
NOTE 2

TRANSFORMER PROTECTION ALARM & TRIP CONTACTS (NOTE 6)

6 6 6 6 TEST 6 6 6 6

6 6 TEST 6 6

110V AC BUS VOLTAGE SUPPLY FROM INCOMER INC-1 BUS VT (PANEL B05)

CONT'D ON SHEET 2 DRG 2000-F00-SLD-EL-10003

TRANSFORMER PROTECTION DEVICES

F13TR402-JB
BALL MILL VSD TRANSFORMER JUNCTION BOX

3x2 3x2 3x2 3x2 3x2 3x2

F13-TR-402
33/3.8/3.8kV 33/3.8/3.8kV
33/0.415/0.415kV
Dyd Dyd Yyd - (NOTE 10)
TOTAL : 25400kVA
PRIMARY : 12000kVA
: 12000kVA
: 1400kVA
SECONDARY : 2x6000 kVA
: 2x6000 kVA
2x700 kVA
KNAN

CT's
1200/1A
5VA
SP20

3x2 3 3x2 3 3x2 3 3x2 3 3x2 3 3x2 3

TRANSFORMER SECONDARY CT's

SWITCHROOM E-SHUT (REFER NOTE 4)
CIRCUIT BREAKER TRIP COMMAND
CIRCUIT BREAKER CLOSE COMMAND
DISCONNECTOR STATUS SIGNAL
CIRCUIT BREAKER CLOSED SIGNAL
110V AC BUS VOLTAGE SUPPLY

ACTIVATION OF SWITCHROOM E-SHUT (REFER NOTE 4)
DETECTION OF ARC FAULT (REFER NOTE 5)

F13-TR-014
RECYCLE CRUSHING MCC 33/0.433kV TRANSFORMER

F13-SW-014
RECYCLE CRUSHING 415V MCC INCOMER

REFER DRG 2000-F13-SLD-EL-10401

F13-TR-003
BALL MILL MCC 33/0.433kV TRANSFORMER

F13-SW-003
BALL MILL 415V MCC INCOMER

REFER DRG 2000-F13-SLD-EL-10901

F13MB401A-VS
BALL MILL MOTOR 1 HV VSD (MASTER) 3.8/3.5kV

F13MB401B-VS
BALL MILL MOTOR 2 HV VSD (SLAVE) 3.8/3.5kV

BALL MILL MOTOR 1 EXCITER PANEL
BALL MILL MOTOR 2 EXCITER PANEL
BALL MILL VSD (MASTER) AUXILIARY PANEL

F13MB401AVS-ISL03 F13MB401AVS-ISL01 F13MB401AVS-ISL02 F13MB401BVS-ISL01 F13MB401BVS-ISL02 F13MB401BVS-ISL03

ISOLATOR PANEL
ISOLATOR PANEL A
ISOLATOR PANEL B
ISOLATOR PANEL A
ISOLATOR PANEL B
ISOLATOR PANEL

CONT'D FROM SHEET 2, DRG 2000-F00-SLD-EL-10003

F13-SR-003
SAG & BALL MILLS VSD SWITCHROOM

EXCITER M 3~ RTD RTD M 3~ EXCITER
10000 kW 10000 kW

F13-MB-401A
BALL MILL MOTOR 1
2000-ME001.3-D02-50007

F13-MB-401B
BALL MILL MOTOR 2
2000-ME001.3-D02-50007

33kV SPARE (1)
FULLY EQUIPPED
-

NOTES

1. THE FEEDER PROTECTION RELAY VOLTAGE INPUT SHALL BE WIRED FROM THE CORRESPONDING INCOMER BUS VT.
2. ARC FLASH POINT DETECTORS SHALL BE PLACED IN CABLE ZONES AND CONNECTED DIRECTLY TO THE SEL PROTECTION RELAYS IN EACH PANEL.
3. SWITCHBOARD SHALL BE EXTENSIBLE FOR FUTURE EXPANSION AND SUPPLIED WITH ONE SPARE TIER PER BUS SECTION AS SHOWN.
4. TRANSFORMER 33kV FEEDER SHALL BE TRIPPED ON THE ACTIVATION OF THE SWITCHROOM EMERGENCY SHUTDOWN (E-SHUT) PUSHBUTTON VIA A CONTACT HARDWIRED DIRECT TO SHUNT TRIP THE CIRCUIT BREAKER.
5. TRANSFORMER 33kV FEEDER SHALL BE TRIPPED ON THE ARC FAULT DETECTION OF THE DOWNSTREAM MCC INCOMER ACB.
6. TRANSFORMER INSTRUMENTATION ALARM & TRIP CONTACTS SHALL BE HARDWIRED TO THE SEL787 TRANSFORMER FEEDER PROTECTION RELAYS.
7. THE INTERTRIP OF THE TRANSFORMER 33kV FEEDER SHALL BE HARDWIRED FROM THE 11kV SWITCHBOARD INCOMER PROTECTION RELAY EITHER ON A TRANSFORMER TRIP, EARTH FAULT DETECTION OR ARC FAULT DETECTION.
8. THE ACTIVATION OF THE PLANT 33kV DISTRIBUTION SWITCHROOM F00-SR-001 EMERGENCY SHUTDOWN (E-SHUT) PUSHBUTTON VIA A CONTACT HARDWIRED DIRECT TO SHUNT TRIP ALL THREE INCOMER CIRCUIT BREAKERS AND THE CORRESPONDING UPSTREAM POLE MOUNTED RECLOSERS.
9. PROTECTION TRIP FROM THE INCOMER CIRCUIT BREAKER SHALL INTERTRIP THE UPSTREAM POLE MOUNTED RECLOSER.
10. REFER TO NAMEPLATE DRAWING 2000-ME001.3-D05-50001 FOR LV WINDING PHASE SHIFT ANGLES.

LEGEND

PR	INCOMER / FEEDER PROTECTION RELAY NOTES 1 & 2
PQM	POWER QUALITY AND REVENUE METERING
	ARC FLASH SENSOR (POINT TYPE) NOTE 2
	CAPACITIVE VOLTAGE INDICATION
	MECHANICAL OR ELECTRICAL INTERLOCK DEVICE
27	UNDER VOLTAGE
50	INSTANTANEOUS OVERCURRENT
50G 50N	INSTANTANEOUS EARTH/NEUTRAL OVERCURRENT
51	TIMED OVERCURRENT
51G 51N	TIMED EARTH/NEUTRAL OVERCURRENT
59	OVER VOLTAGE
64	EARTH FAULT
67 67G	DIRECTIONAL OVERCURRENT/EARTH FAULT
AFD	ARC FLASH DETECTION

2000-F00-SLD-EL-10002_Rev5

Single line diagram sheet 1